



# ON-BOARD DIAGNOSTIC REPORT

Work Order WO-2026-0406-01 • Injector centering / quantity correction read

## 01 VEHICLE

| Field      | Value                                        |
|------------|----------------------------------------------|
| VIN        | JMBXLGA1WEZ006344                            |
| Make       | MITSUBISHI                                   |
| Model      | ASX                                          |
| Year       | 2014                                         |
| Engine     | 4N13 1.8 DiD (1798 cc, diesel, common-rail)  |
| Drivetrain | AWD                                          |
| ECU map    | MMC_ASX_4N13                                 |
| CAN bus    | 500 kbps • TX 0x7E0 • RX 0x7E8 (ISO 15765-4) |

## 02 DIAGNOSTIC ADAPTER

| Field      | Value                                           |
|------------|-------------------------------------------------|
| Make/Model | Autocom CDP+ (USB FTDI 0403:d6da)               |
| MCU        | STM32F205ZGT6 + FT232R                          |
| Firmware   | 1622                                            |
| Serial     | 100251                                          |
| Battery V  | 13.62 V (engine off) ? 13.66 V (running)        |
| Protocol   | ASCII text wrapper • CAN/ISO 15765-4 underneath |

## 03 ECU PARAMETER MAP

56 unique parameter indices discovered via \*608\_21\_&lt;XX&gt; read commands.

Indices grouped by purpose (preliminary classification, pending operator validation):

| Range | Count | Preliminary classification                        |
|-------|-------|---------------------------------------------------|
| 01-19 | 16    | Core ECU sensors (temps, voltages, status flags)  |
| 24-28 | 4     | Auxiliary status / minor counters                 |
| 46-5F | 13    | ASCII fragments likely injector calibration codes |
| 74    | 1     | Single isolated parameter                         |
| A0-A8 | 9     | Calibration lookup table (X?Y pairs, Y constant)  |
| B0-B9 | 10    | Injector quantity correction (10 work points)     |
| BE-BF | 2     | End-of-range telemetry                            |

## 04 INJECTOR QUANTITY CORRECTION (B0..B9)

Decoded as 4-byte field per index = two 16-bit big-endian values per work point.

Pattern matches typical diesel injector centering data (10 work points ? current/target).

| WP | raw1 | raw2 | val1 | val2 | ?   | ? %    |
|----|------|------|------|------|-----|--------|
| B0 | 01EB | 01E2 | 491  | 482  | -9  | -1.8 % |
| B1 | 01DF | 01D5 | 479  | 469  | -10 | -2.1 % |
| B2 | 01DA | 01DA | 474  | 474  | 0   | 0.0 %  |
| B3 | 01D1 | 01CE | 465  | 462  | -3  | -0.6 % |
| B4 | 01D0 | 01CA | 464  | 458  | -6  | -1.3 % |
| B5 | 01D3 | 01C8 | 467  | 456  | -11 | -2.4 % |
| B6 | 01CA | 01CB | 458  | 459  | +1  | +0.2 % |
| B7 | 01D4 | 01CE | 468  | 462  | -6  | -1.3 % |
| B8 | 01B8 | 01C4 | 440  | 452  | +12 | +2.7 % |
| B9 | 01C9 | 01C2 | 457  | 450  | -7  | -1.5 % |

? = val2 - val1 (target ? current). Pattern shows minor drift, all within typical -20 unit tolerance.



05 OBSERVATIONS & ACTIONS

- ? No fault codes (DTC) read during this session diagnostic was parameter-only.
- ? ECU responds to all 56 polled parameter indices without timeout.
- ? Injector correction values within tolerance, no replacement required.
- ? 8 parameter responses were truncated by sniffing layer (not ECU side);  
a follow-up capture using full-payload sniffer is scheduled.
- ? Adapter handshake clean: device name CDP+, fw 1622, S/N 100251.
- ? No unsafe writes performed. Read-only session.

06 TECHNICIAN SIGN-OFF

|             |                                                               |
|-------------|---------------------------------------------------------------|
| Technician: | smooker                                                       |
| Role:       | Diagnostic Technician                                         |
| Date:       | 2026-04-06T18:46                                              |
| Work Order: | WO-2026-0406-01                                               |
| Signature:  | <div><i>smooker</i><br/>/digital signature placeholder/</div> |

DRAFT